

Objective:

The primary objective of CartSense is to develop a modular, AI-assisted shopping system that simplifies product identification, cart management, and checkout in retail environments. The system aims to use a customer's compatible smartphone as the primary camera, processing interface, and display, enabling supported products to be identified using computer vision and maintained in a real-time digital cart.

The system will maintain product names, quantities, prices, and the running cart total, while providing mechanisms to add or remove items and generate a final bill. The solution will focus initially on a predefined catalogue of packaged products and will be designed to support recognition of multiple products where technically feasible.

CartSense will explore the integration of computer vision, deep learning, mobile/web technologies, backend services, and optional IoT or cart-connectivity mechanisms. The system will follow a modular architecture so that it can be adapted to existing shopping carts and extended with additional products, sensing technologies, inventory integration, and retail analytics in future versions.

The project aims to demonstrate a practical alternative to conventional manual product scanning, with the potential to reduce checkout time and provide customers with greater visibility of their purchases while retaining the conventional checkout process as a fallback.

Description:

CartSense is a smartphone-assisted automated shopping system designed to make retail shopping and checkout more convenient. A customer can connect their compatible smartphone to a CartSense-enabled shopping session, using an appropriate connection mechanism such as a QR code or other supported method.

During shopping, the smartphone camera captures the products selected by the customer. A computer vision model processes the captured visual information and identifies products from a predefined catalogue. Where supported, multiple products can be detected within the camera view. The identified products are then added to a digital cart, which maintains their names, quantities, prices, and running total.

The customer can view the cart and its updated bill while shopping and can remove items when required. At the end of the session, the system can generate or present the final bill through the checkout interface.

The initial implementation will focus on a controlled set of packaged products with distinguishable visual characteristics. The system is designed as a modular foundation that can subsequently incorporate improved product recognition, barcode-based fallback mechanisms, additional sensing, inventory synchronisation, retailer-side management, and analytics.

CartSense is intended to demonstrate how AI, mobile computing, and modular retail technology can work together to reduce manual product scanning and streamline the shopping-to-checkout workflow without requiring complete replacement of existing shopping infrastructure.